# 1. Frobenius-orbit index calculus on Koblitz curves, fully priced, against the best rho

2026-10-14 · Conductor T-1 · Class: **engineering** · Study `research/ecbench_frobenius_orbit_vs_rho_20261014` · Protocol frozen in commit `e1708eef8` before claim data.

**Verdict: constant-only.** With every native priced, Frobenius-orbit IC ($m = 2$, `compact-orbit-scan` + `mitm-frobenius-counted`, `stride-renew` targets) is not below rho-frob on any of eight Koblitz curves ($\log_2 r$ = 11–39). The paired ratio grows with $r$: slope $\beta = 0.165[0.117, 0.213]$ per $\ln r$ with $\ln n$ in the fit (model: $\frac{1}{6}$). Orbits are 2–15× cheaper than the subspace base (constant), and beat rho-strong only where rho-strong's fixed set-up dominates ($\log_2 r \leq 17$). No security claim.

## 1.1. What was built

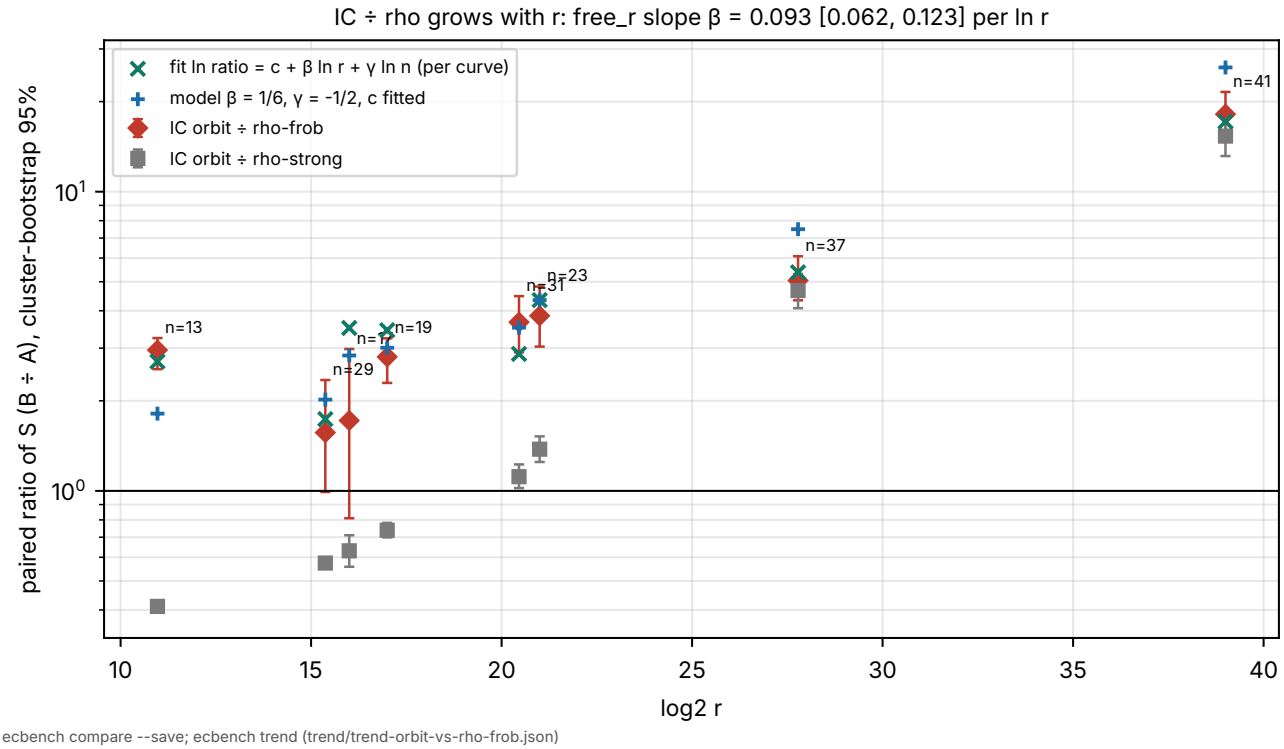
- **Full pricing.** New opt-in methods `ic.pipeline_full`, `rho.signed_frobenius_full`, `rho.signed_frobenius_strong_full` run unchanged solvers, then price every native counter from `docs/ecbench/full_prices.json` (ratios to one affine addition measured by `ecbench price-units`, median of 15 passes, this host). Unknown counters stay unpriced. Existing methods and `docs/ic/calibration.json` unchanged; committed sessions replay as before.
- **`ecbench trend`**: weighted least squares of ln(paired ratio) on $\ln r$ and $\ln n$ from `compare --save` files.

## 1.2. Panel and verification

Eight sessions, 16 fresh public targets × 3 rounds (+1 warm-up), L0, 4 vCPU Xeon 2.80 GHz shared. 2,816/2,816 records verified; 2,112/2,112 measured records replayed identically; A/A exactly 1.000 on every curve; no record a lower bound.

| curve | $\log_2 r$ | h | c | orbit ÷ rho-frob | orbit ÷ rho-strong | renew ÷ walk | orbit ÷ negfold |
|---|---|---|---|---|---|---|---|
| a0 n13 | 10.97 | 4 | 1 | 2.956 [2.552, 3.242] | 0.411 [0.397, 0.423] | 0.403 [0.395, 0.412] | 0.511 [0.465, 0.562] |
| a1 n17 | 16.00 | 2 | 1 | 1.719 [0.810, 2.980] | 0.631 [0.558, 0.711] | 0.503 [0.448, 0.562] | 0.234 [0.203, 0.270] |
| a0 n19 | 17.00 | 4 | 2 | 2.805 [2.295, 3.236] | 0.739 [0.699, 0.782] | 0.556 [0.525, 0.591] | 0.148 [0.131, 0.170] |
| a0 n23 | 21.00 | 4 | 3 | 3.847 [3.035, 4.819] | 1.379 [1.249, 1.521] | 0.753 [0.678, 0.834] | 0.065 [0.059, 0.071] |
| a1 n29 | 15.37 | 12646 | 1 | 1.566 [0.992, 2.347] | 0.574 [0.556, 0.593] | 0.480 [0.464, 0.499] | – |
| a0 n31 | 20.46 | 1492 | 2 | 3.664 [2.907, 4.476] | 1.116 [1.022, 1.226] | 0.668 [0.607, 0.733] | – |
| a0 n37 | 27.78 | 596 | 8 | 5.048 [4.335, 6.086] | 4.676 [4.080, 5.310] | 1.029 [0.903, 1.176] | – |
| a0 n41 | 39.00 | 4 | 90 | 18.16 [15.67, 21.55] | 15.34 [13.16, 18.28] | 1.009 [0.938, 1.082] | – |

Paired B ÷ A ratios of S from `ecbench compare`, cluster-bootstrap 95 % intervals. $h$ = cofactor, $c$ = orbit columns.



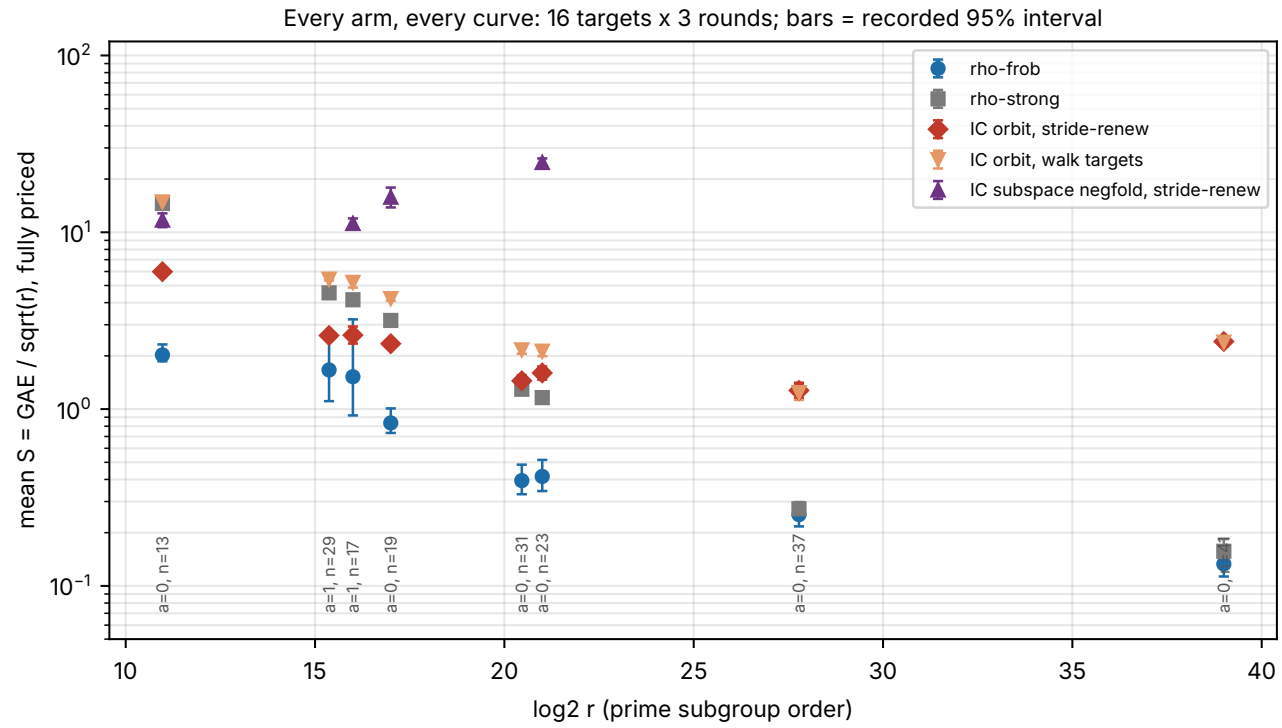

ecbench compare --save; ecbench trend (trend/trend-orbit-vs-rho-frob.json)

Figure 1: Paired ratios against $\log_2 r$ with the `free_r_n` and fixed-model fitted values. Sessions as listed in README; `trend/trend-orbit-vs-rho-frob.json`.

### 1.3. Trend (H6)

| fit | coefficients (95 %) | $\chi^2$/dof |
|---|---|---|
| $c + \beta \ln r$ | $\beta = 0.093[0.062, 0.123]$ | 9.9 |
| $c + \beta \ln r + \gamma \ln n$ | $\beta = 0.165[0.117, 0.213]$, $\gamma = -1.18[-1.90, -0.46]$ | 3.9 |
| model $c + \ln r/6 - \ln n/2$ | $e^c = 1.84[1.41, 2.40]$ | 15.7 |

$\beta > 0$ in both free fits: constant-only by the frozen rule. The fixed model fails the $\chi^2$/dof ≤ 2 rule; the free $\gamma$ rests on the two large-cofactor small-$r$ curves where $c$ is at its integer floor and rho-frob is heavy-tailed, so it is descriptive. Extrapolating `free_r_n` to $r \approx 2^n$ (derived, not measured): the ratio is smallest near $n \approx 10$ at about 3.3 and never below 1.

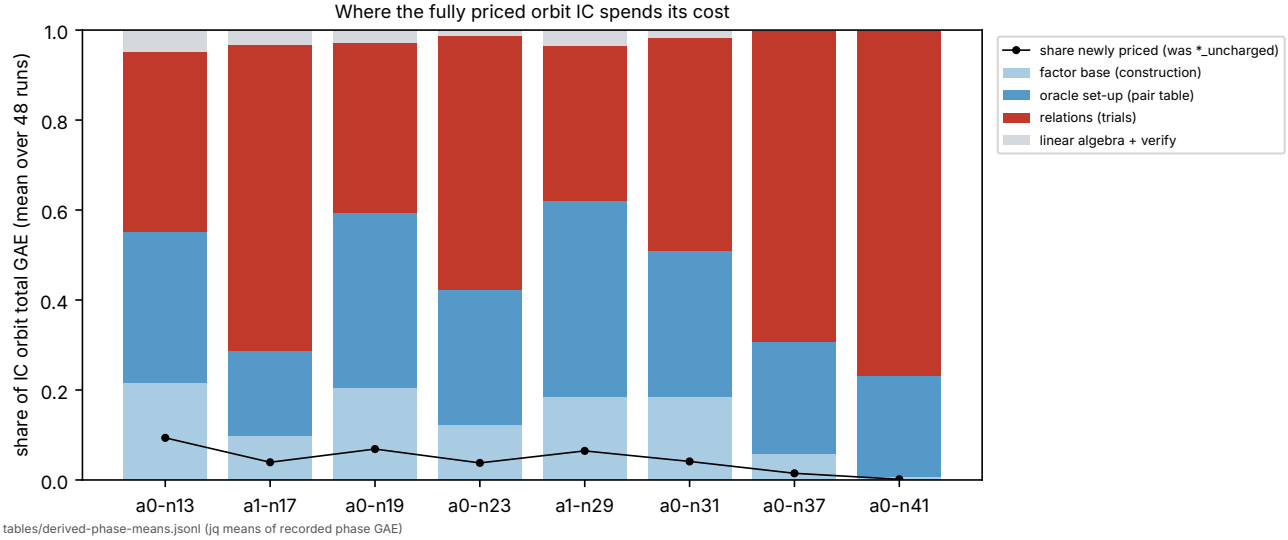



Sessions: research/ecbench_frobenius_orbit_vs_rho_20261014/sessions/*; ecbench table

Figure 2: Mean S of every arm on every curve with recorded 95 % intervals (`ecbench table`).

### 1.4. Pricing

The orbit base's formerly `*_uncharged` construction work costs 9.4 %, 6.9 %, 3.8 % and 0.16 % of the IC total at n = 13, 19, 23, 41: the old lower bound hid little. All natives are 36 % (n = 13) to 16 % (n = 41) of the IC total; hash probes are priced on a $2^{20}$-entry map, about 11× the pinned ratio, which leans against IC. With every native priced at zero on both sides (derived, ratio of means), IC ÷ rho-frob is 1.98, 1.28, 1.84, 2.77, 0.96, 2.60, 4.09, 16.3 in panel order, so the verdict does not hinge on prices.




tables/derived-phase-means.jsonl (jq means of recorded phase GAE)

Figure 3: Phase shares of the orbit-renew arm (means of recorded phase GAE, `tables/derived-phase-means.jsonl`).

### 1.5. Limits

Toy and medium sizes on one shared host (L0; wall descriptive); prices are one host's microbenchmark ratios; $c$ and $d$ chosen on exploratory seeds (disclosed); negfold only at n ≤ 23 with cofactor ≤ 4; rho exponents fitted on these sizes are pre-asymptotic (rho-frob α = 0.238 [0.135, 0.303]); two a1 sessions record `git_dirty` from a whitespace-only `rustfmt` during the run (binary built from `e1708eef8`, unchanged).